Photocatalyst-DFT

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Abstract

A reproducible Python pipeline was built (RDKit, PySCF, geomeTRIC) to describe the electronic structure of 16 reported organic photocatalysts, covering xanthene dyes, aryl ketones, acridinium, phenothiazine, perylene diimide, porphyrin and donor–acceptor systems. Each molecule's identity was checked against its catalogue formula, charge and InChIKey. Three-dimensional structures were generated by ETKDG conformer search with MMFF94s relaxation. Six molecules were then re-optimised with B3LYP/def2-SVP in implicit water (ddCOSMO), and frontier-orbital energies were computed for all 16 at the same level. The Kohn–Sham HOMO–LUMO gap tracks the lowest experimental absorption band well ($r = 0.89$) but overestimates it by 0.46 eV on average. This is expected, because orbital gaps neglect electron–hole attraction; benzophenone, whose lowest band is $n \to \pi^*$, is the main outlier. Mulliken analysis on molecular fragments shows that the frontier orbitals of Mes-Acr$^+$ sit almost entirely on separate, orthogonal donor and acceptor rings, while both frontier orbitals of eosin Y^{2-} lie on its xanthene core. Comparing MMFF and DFT geometries gives an error bar of ≤ 0.03 eV in the gap for rigid molecules and up to 0.15 eV for flexible or charged dyes. Orbital energies are reported only as a qualitative ranking, not as redox potentials. The next steps are time-dependent DFT excitation energies, dispersion-corrected geometries and extension of the survey to solid-state photocatalysts.

Current Status The organic half of the project is built, verified and documented. Sixteen organic photocatalysts are catalogued, cross-checked against PubChem, embedded in 3D and given B3LYP/def2-SVP frontier orbitals in implicit water. Six of them have full DFT geometry optimisations. Step 5 is now complete. It shows that MMFF geometries are adequate for rigid dyes ($|\Delta\text{gap}| \leq 0.03$ eV) but not for flexible or charged ones (up to 0.15 eV, eosin Y^{2-}). That is about a third of the 0.46 eV scatter of the gap against experiment ($r = 0.89$). Next are DFT geometries for the remaining 10 dyes and TDDFT. This summary covers the organic part only.

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1 Project map

The pipeline is five scripts. Each one reads the previous step's files and writes its own, so any step can be rerun on its own.

Step	Script	What it does → main output
1	`01_represent.py`	SMILES → checks → multi-conformer 3D (ETKDGv3 + MMFF94s) → `molecules/xyz/*.xyz`, 2D grid, 3D HTML viewer
2	`02_optimize.py`	DFT geometry optimisation (PySCF + geomeTRIC) → `molecules/xyz_opt/*.xyz`, `geomopt_summary.csv`
3	`03_frontier.py`	DFT single point on the best available geometry → HOMO/LUMO for all 16, `frontier_orbitals.csv`, energy-level diagram
4	`04_orbitals.py`	Orbital cube files + fragment localisation (Mulliken) → `orbital_localization.csv`, isosurface figure
5	`05_geometry_check.py`	Does the geometry (MMFF vs DFT) change the orbital energies? → `geometry_sensitivity.csv` (6 molecules)
—	`utils/common.py`	Shared code: xyz read/write, building the PySCF molecule, the DFT settings, constants, plotting style
—	`utils/dispersion_shim.py`	**D**3/D4 dispersion backend for PySCF on Apple Silicon (Section **??**)

The level of theory is the same everywhere: **B3LYP/def2-SVP, density fitting, ddCOSMO water ($\varepsilon = 78.36$), restricted closed shell, no dispersion correction.** The stored results are still without it, and Section **??** shows why that changes little for rigid molecules.

2 Step 1 — Catalogue and molecular representation

2.1 The catalogue

`data/organic_photocatalysts.csv` has one row per molecule and 20 columns. The ones the code depends on are `slug` (the file-name ID used by every later step), `smiles`, `formula`, `charge` and `inchikey`. The rest is literature metadata: family, role, $\lambda_{\max}$, the lowest absorption band and redox notes.

PubChem cross-check. Each name was resolved against PubChem. 9 of 16 stored species match a PubChem record exactly. For the other 7, `pubchem_parent_cid` points to a *related* record, and `pubchem_cid_relation` explains how they relate:

- salts (EY, RB, RhB, MB, Mes-Acr): PubChem stores the salt; we keep only the active ion;
- fluorescein: PubChem's main record is the neutral lactone; we use the quinoid dianion;
- Me-PDI: no PubChem record; built from the parent PTCDA (CID 67191).

Ions: Compute the species that does the chemistry. In water at working pH, eosin Y is a dianion, and its colour and redox potentials belong to that dianion. The InChIKey records this in its last letter: neutral eosin Y ends in `-N`, the dianion in `-L` (two protons removed). The first 14

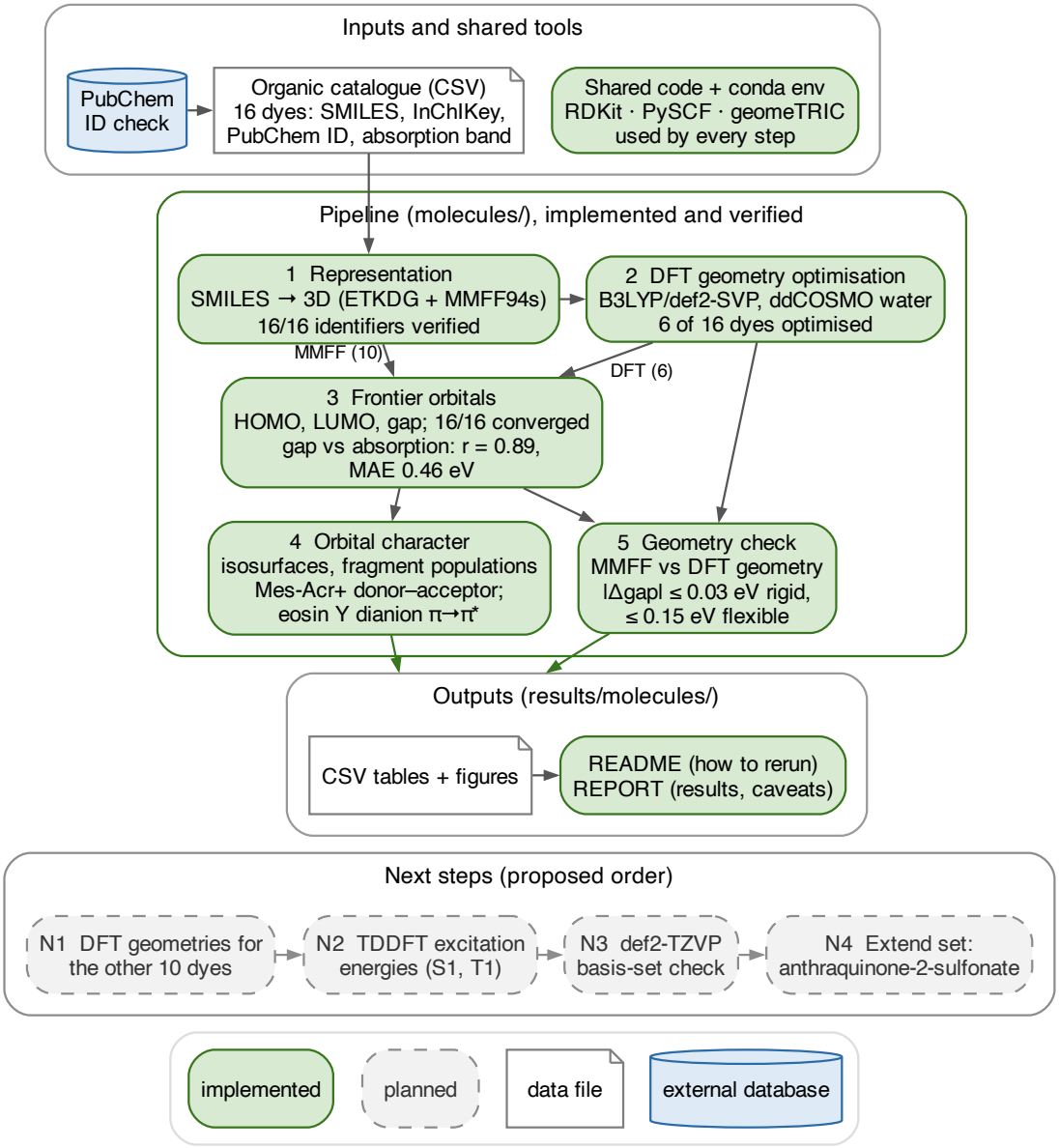


Figure 1: Code architecture and development plan (organic part). Green: implemented and verified by rerunning. Grey: proposed next steps, in order.

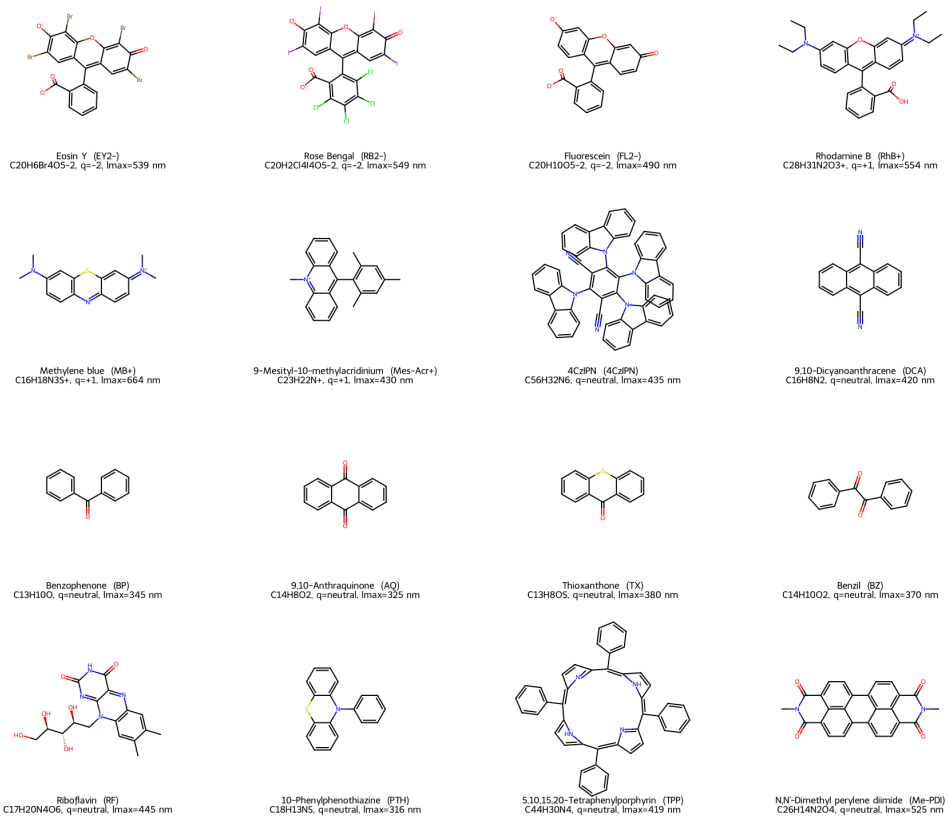


Figure 2: The 16 molecules as encoded. Charges are part of the identity: EY^{2-}, RB^{2-} and FL^{2-} are dianions; MB^{+}, RhB^{+} and Mes-Acr^{+} are cations.

characters, which encode the skeleton, are identical. That is why a salt or ion can be traced back to its parent record in PubChem.

2.2 Checks before any 3D work

For each row the script parses the SMILES (RDKit returns `None` on a syntax error; the molecule is then skipped). It then computes formula, total charge and InChIKey from the parsed structure and compares each with the stored value. Mismatches go into a `problems` list instead of stopping the run, so all errors are reported at once.

Caveat: What the InChIKey check can and cannot catch. The formula and charge checks catch real typos. For example, dropping one `(Br)` from the eosin Y SMILES still gives a valid SMILES, but the formula becomes $C_{20}H_7Br_3O_5{}^{2-}$. The InChIKey check, however, can only catch *later edits*, because the stored key was itself generated from the same SMILES. The independent check is the PubChem comparison above.

2.3 Multi-conformer 3D embedding

1. `AddHs`: implicit hydrogens become real atoms, so they get coordinates (eosin Y: 29 → 35 atoms).
2. **ETKDGv3** distance-geometry embedding with a fixed random seed (`0xF00D`). It generates 8 conformers, or 30 if the molecule has more than 3 rotatable bonds. Conformers within 0.5 Å RMSD of each other are pruned as duplicates.
3. Every conformer is minimised with **MMFF94s**, and the lowest-energy one is written to xyz with `charge=... mult=1` in the comment line. The xyz format has no charge field; `read_xyz`

recovers it with the regular expression `charge=(-?\d+)`.

Molecule	Rotatable bonds	Conformers tried	Kept after pruning
anthraquinone	0	8	1
eosin Y^{2-}	2	8	2 (mirror images, identical energy)
riboflavin	9	30	18

Concept: Conformer count as a measure of flexibility. The number of surviving conformers measures flexibility. A rigid molecule collapses to one structure however many times you embed it. Riboflavin's ribityl chain gives 18 distinct shapes spread over 14 kcal/mol in MMFF energy. The two eosin Y conformers have the benzoate ring twisted one way or the other, which is why they are exactly degenerate.

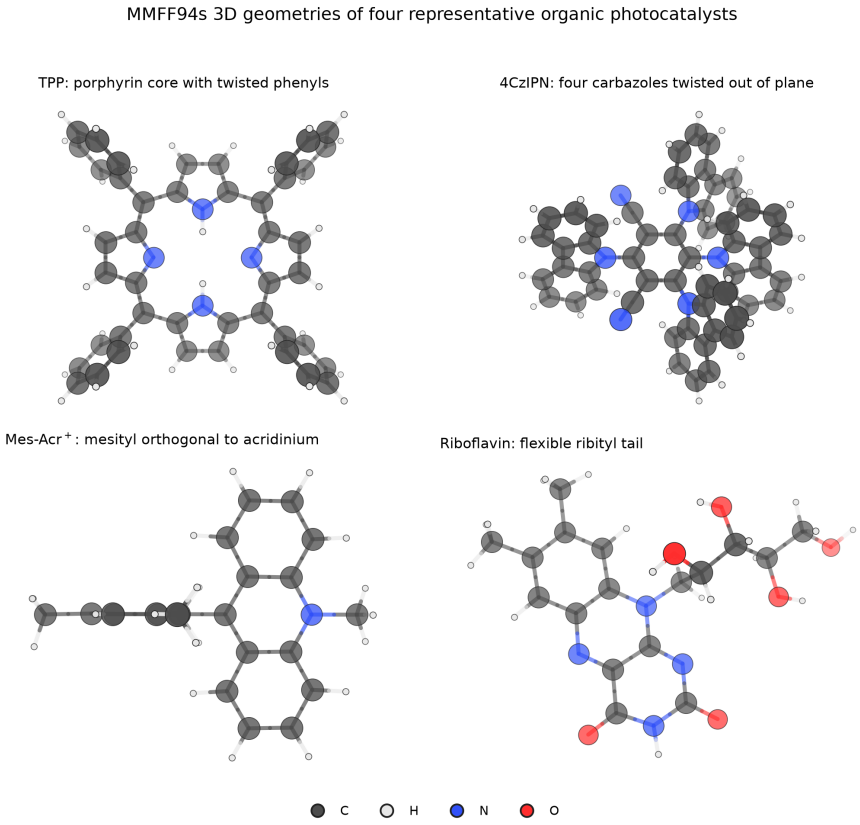


Figure 3: MMFF94s geometries. Each feature shown is a twist that decouples two π systems (TPP phenyls, 4CzIPN carbazoles, Mes-Acr^+ mesityl) or a flexible tail (riboflavin).

3 Step 2 — DFT geometry optimisation

3.1 What the script does

1. Builds the PySCF molecule from the MMFF xyz (`build_mol`). def2 effective core potentials are added automatically for elements heavier than Kr ($Z > 36$), i.e. iodine in rose bengal; bromine ($Z = 35$) does not need one.
2. Runs an SCF on the MMFF geometry, which gives the reference energy $E_{\text{MMFF geom}}$.
3. Optimises with **geomeTRIC**, which steps in internal coordinates (bonds, angles, torsions) rather than raw x, y, z.

4. Runs a final SCF at the optimised geometry, then compares the two structures with a **Kabsch-aligned RMSD**.

Concept: Kabsch alignment. Two copies of the same molecule can differ simply by being rotated or shifted. Kabsch's algorithm first centres both copies, then finds the rotation that best superimposes them (via a singular value decomposition of $P^{\mathsf{T}}Q$). Only then does it measure the per-atom distances. Without it, an RMSD would mix real structural change with an arbitrary orientation. The script reports RMSD over all atoms, over heavy atoms only, and the single largest atomic displacement.

3.2 Convergence criteria: `gau` vs `flat`

An optimisation stops when *all* of its tests pass. The default (`gau`, Gaussian-like) has three kinds of test:

Test	`gau`	`flat`
energy change per step	10^{-6} Ha	10^{-5} Ha
gradient (RMS / max)	3×10^{-4} / 4.5×10^{-4} Ha/bohr	same
displacement (RMS / max)	1.2×10^{-3} / 1.8×10^{-3} bohr	effectively off

For eosin Y, 27 `gau` steps met the gradient test, but the displacement test never settled. The optimiser kept taking sizeable steps along very soft torsions (the benzoate twist) at almost no cost in energy. The run was therefore resumed from its last frame with `--criteria flat`, which keeps the same gradient thresholds but drops the displacement test. It converged in 4 more steps. Riboflavin and rhodamine B used `flat` from the start.

Concept: Flat potential energy surfaces. The gradient tells you the surface is level; the displacement tells you the optimiser has stopped moving. On a very flat torsional surface the gradient can be small while the atoms still wander. Relaxing the displacement test is reasonable *because* the gradient test still holds. The structure is a true stationary point, but the exact torsion angle is poorly determined. Report it with that caveat.

3.3 Results

Molecule	AO	Cycles	ΔE (kcal/mol)	$\mathrm{RMSD}_{\mathrm{heavy}}$ (Å)	Max dev. (Å)	Criteria
thioxanthone	254	10	−1.88	0.030	0.06	gau
anthraquinone	264	15	−1.42	0.019	0.04	gau
benzophenone	246	11	−1.27	0.096	0.26	gau
riboflavin	478	18	−11.13	0.173	0.65	flat
eosin Y^{2-}	508	31	−17.73	0.316	1.30	gau → flat
rhodamine B^{+}	617	20	−20.23	0.398	1.27	flat

$\Delta E = E(\text{DFT-optimised}) - E(\text{MMFF geometry})$, both at the same DFT level. Recorded wall times: rhodamine B 68 073 s (≈19 h), riboflavin 5 547 s (≈1.5 h); eosin Y's recorded 711 s covers only the 4-step resume. The rhodamine B figure is elapsed time and includes about 15.5 h during which the laptop was asleep; the optimisation itself needs about 2 h (Section 7.1).

Key finding: MMFF geometries are not adequate for every dye. The earlier conclusion "MMFF is good enough, so skip DFT for the other 13" does not hold. It was based on three rigid molecules. The flexible and charged dyes moved roughly 6–13 times more than the rigid ones (heavy-atom RMSD 0.17–0.40 vs 0.02–0.03 Å) and dropped 11–20 kcal/mol in energy.

What actually moved. I aligned each MMFF and DFT pair and listed the atoms that moved most:

- **Eosin Y^{2-}**: the largest mover is a carboxylate oxygen (1.30 Å). The ring–ring twist changed from 98.6° to 86.1°.
- **Rhodamine B^{+}**: again a carboxylic-acid oxygen (1.27 Å), with the ring–ring twist almost unchanged ($-84.9° \to -88.9°$). The next movers are ethyl hydrogens.
- **Riboflavin**: hydroxyl hydrogens and oxygens of the ribityl chain (up to 0.65 Å). This looks like a hydrogen-bond rearrangement.

So the large RMSDs come mainly from the **carboxyl groups and side chains rotating**, not from the chromophore cores. A likely reason is that MMFF minimised these ions without any solvent, while the DFT run used ddCOSMO water. A bare $-COO-$ group in vacuum and a solvated one prefer different orientations. Because the HOMO and LUMO of these dyes sit on the cores, the orbital energies may change much less than the RMSD suggests. Step 5 tested this (Section 6). The gap moves by up to 0.15 eV: smaller than the RMSD might suggest, but not negligible.

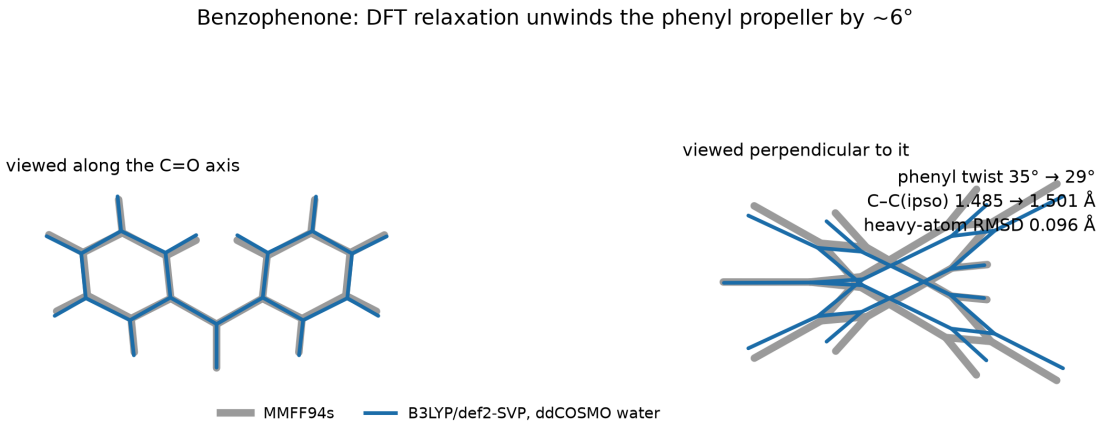


Figure 4: Benzophenone, MMFF94s (grey) vs DFT (blue): the change is invisible along C=O and obvious from the side. Always look from two directions.

4 Step 3 — Frontier orbital energies

A single point on each molecule's best available geometry: the DFT-optimised one if it exists, otherwise the MMFF one. The column `geom_source` records which was used. If the SCF fails to converge, the script retries with a second-order (Newton) solver. All 16 converged.

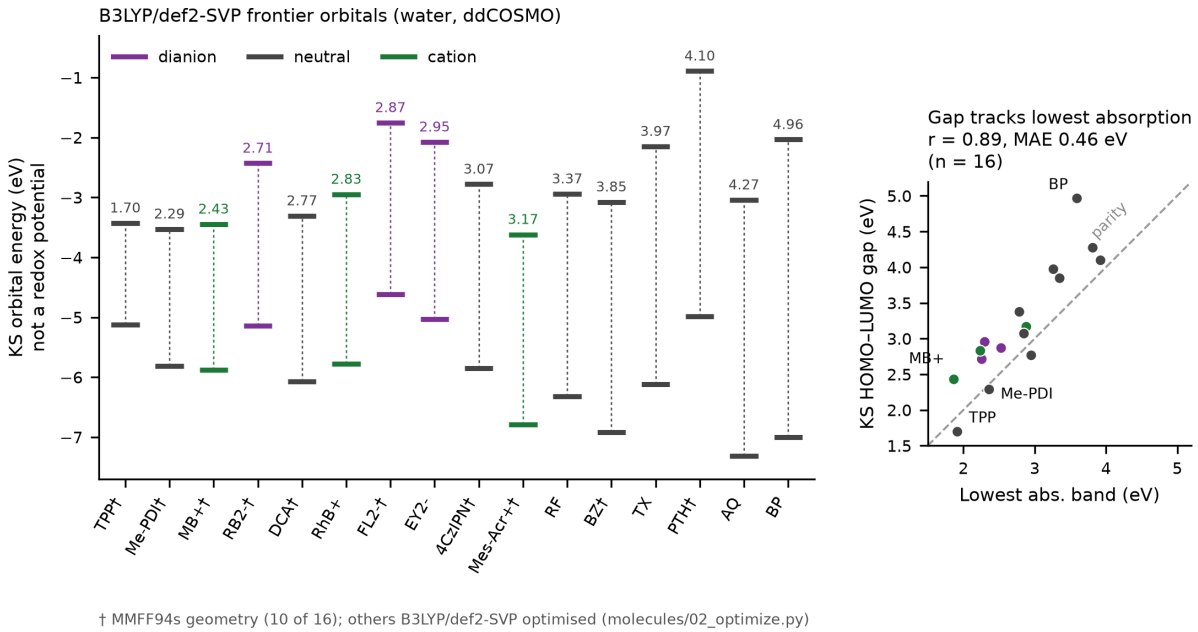


Figure 5: Current diagram. Compared with the first version, the H^+/H_2 and O_2/H_2O lines were removed, the axis now says "not a redox potential", and † marks the 10 molecules still on MMFF geometry.

PC	HOMO (eV)	LUMO (eV)	Gap (eV)	Lowest abs. (eV)	Geometry
TPP	−5.12	−3.43	1.70	1.92	MMFF
Me-PDI	−5.81	−3.53	2.29	2.36	MMFF
MB^+	−5.87	−3.45	2.43	1.87	MMFF
RB^{2-}	−5.14	−2.43	2.71	2.26	MMFF
DCA	−6.07	−3.31	2.77	2.95	MMFF
RhB^+	−5.78	−2.95	2.83	2.24	DFT
FL^{2-}	−4.62	−1.75	2.87	2.53	MMFF
EY^{2-}	−5.03	−2.07	2.95	2.30	DFT
4CzIPN	−5.85	−2.78	3.07	2.85	MMFF
Mes-Acr^+	−6.79	−3.62	3.17	2.88	MMFF
RF	−6.32	−2.94	3.37	2.79	DFT
BZ	−6.92	−3.08	3.85	3.35	MMFF
TX	−6.12	−2.15	3.97	3.26	DFT
PTH	−4.99	−0.89	4.10	3.92	MMFF
AQ	−7.31	−3.04	4.27	3.81	DFT
BP	−7.00	−2.03	4.96	3.59	DFT

Key finding: Agreement with experiment. The Kohn–Sham gap tracks the lowest experimental absorption band ($r = 0.89$, MAE 0.46 eV, $n = 16$), and is larger than it for 13 of the 16. (Before eosin Y, riboflavin and rhodamine B were moved onto their DFT geometries, these were $r = 0.90$ and MAE 0.45 eV.) The largest outlier is benzophenone (+1.37 eV), whose weak lowest band is n→ π^*. TPP is compared against its $Q_x(0,0)$ band (647 nm), not the Soret band, which is S_2.

Concept: Three gaps that are not the same thing.

- *KS HOMO–LUMO gap*: a difference of orbital energies from one ground-state calculation.
- *Fundamental gap* = IE − EA: needs calculations on the cation and anion. For benzene, B3LYP's $-\varepsilon_{\mathrm{HOMO}}$ was 6.97 eV against an experimental IE of 9.24 eV.
- *Optical gap*: the lowest excitation, which includes the electron–hole attraction. It needs TDDFT.

A good correlation with absorption says the *trend* is captured, not that the absolute values mean anything. This is why the redox lines were removed from the diagram.

5 Step 4 — Where the orbitals live

For Mes-Acr$^+$ and eosin Y^{2-}, the script writes cube files for HOMO−2 to LUMO+1 (about 4.5 MB each, git-ignored). It then measures how much of each orbital sits on a "core" fragment, which is defined by a SMARTS pattern. The measure is the Mulliken population of orbital ϕ_k on the fragment's atoms:

$$p_k(\text{core}) = \sum_{\mu \in \text{core}} c_{\mu k}\,(\mathbf{S}\mathbf{c}_k)_\mu,$$

where $c_{\mu k}$ are the MO coefficients and $\mathbf{S}$ is the AO overlap matrix. Summed over all atoms this gives exactly 1.

Molecule	Orbital	E (eV)	Localisation
eosin Y^{2-}	HOMO	−5.03	98% xanthene core
	LUMO	−2.08	96% xanthene core
	HOMO−1, HOMO−2	−5.54, −5.88	79%, 50% benzoate
Mes-Acr$^+$	HOMO−2	−6.80	>99% mesityl
	HOMO−1	−6.79	96% mesityl
	HOMO	−6.79	99% acridinium
	LUMO	−3.62	98% acridinium

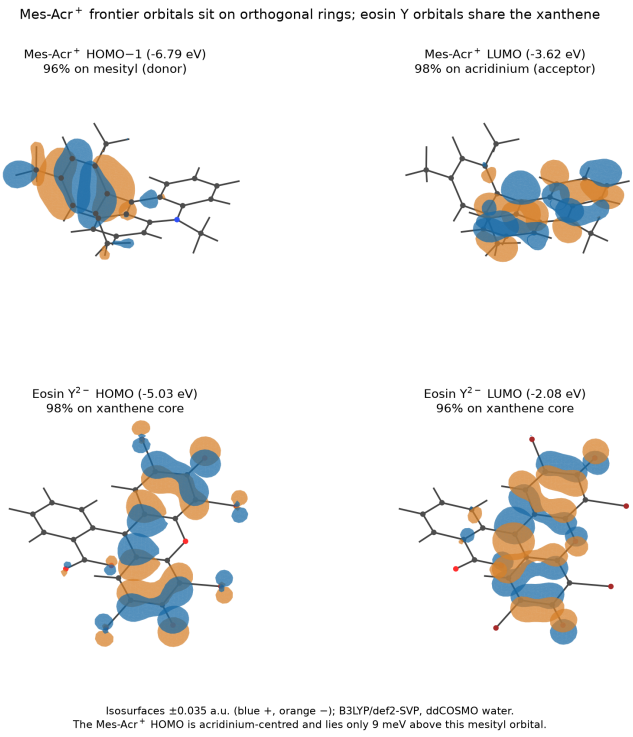


Figure 6: Isosurfaces at ±0.035 a.u., rendered by marching cubes in matplotlib (schematic: matplotlib only approximates which surface is in front).

Key finding: Local versus charge-transfer excitations. Eosin Y's lowest excitation is *local* ($\pi \to \pi^*$ on the xanthene). In Mes-Acr$^+$, donor (mesityl) and acceptor (acridinium) orbitals sit on two orthogonal rings, which gives *charge-transfer* character. This is the orbital-level picture of the 3D twist in Section 2.3. The top three occupied orbitals of Mes-Acr$^+$ lie within 13 meV of each other, so which one is labelled "HOMO" is not meaningful at this level of theory.

Caveat: How to read the orbital percentages. Mulliken populations depend on the basis set. Use them for qualitative statements (91% vs 10%), not for small differences. The eosin Y values are now on its DFT geometry. On the MMFF geometry the HOMO was 91% and the LUMO 97% on the core: the benzoate ring twists further out of conjugation after optimisation, so the HOMO becomes more core-centred. The conclusion ($\pi \rightarrow \pi^*$ on the xanthene) is unchanged.

6 Step 5 — Does the geometry matter for the orbitals?

`05_geometry_check.py` answers the question raised by Section 3.3. For every molecule with a DFT geometry, it reruns the step-3 single point on the *MMFF* geometry and compares it with the step-3 result on the *DFT* geometry (Δ = DFT − MMFF). Step 3 was first rerun for eosin Y, riboflavin and rhodamine B on their DFT geometries, which removed the blocker noted in the previous version.

PC	$\mathrm{RMSD}_{\mathrm{heavy}}$ (Å)	ΔE_{opt} (kcal/mol)	ΔHOMO (eV)	ΔLUMO (eV)	Δgap (eV)
AQ	0.019	−1.4	−0.026	−0.020	+0.006
TX	0.029	−1.9	+0.024	−0.002	−0.027
BP	0.096	−1.3	−0.089	−0.087	+0.001
RF	0.173	−11.1	+0.066	+0.014	−0.052
EY^{2-}	0.316	−17.7	−0.005	+0.143	+0.149
RhB^{+}	0.398	−20.2	−0.069	−0.003	+0.065

Key finding: How much the gap depends on the geometry. For rigid molecules the gap is insensitive to the geometry ($|\Delta\mathrm{gap}| \leq 0.03$ eV), although benzophenone's individual orbitals shift by about 0.09 eV. For the flexible and charged dyes the gap shifts by up to +0.15 eV (eosin Y^{2-}, almost all from its LUMO). That is about a third of the 0.46 eV scatter against experiment. MMFF geometries are therefore good enough for qualitative trends, but not for quantitative comparisons between dyes.

Caveat: Which dyes still need DFT geometries. 10 molecules are still on MMFF geometries: DCA, BZ, PTH, RB^{2-}, FL^{2-}, MB^{+}, Mes-Acr^{+}, Me-PDI, TPP, 4CzIPN. Rose bengal and fluorescein share eosin Y's benzoate–xanthene scaffold and probably carry a similar 0.1–0.15 eV gap error. They are first in line for DFT optimisation.

7 Problems encountered and how they were handled

Problem	Resolution
`pyscf-dispersion` unusable on the Mac (bad library call; x86_64-only binaries)	Replaced by an adapter on the conda-forge D3/D4 libraries; validated; D3 now available
MMFF geometries justified with three rigid molecules only (review comment)	Optimised three flexible/charged dyes and added step 5: gap error up to 0.15 eV, now stated in all outputs
Orbital energies drawn against H^+/H_2 and O_2/H_2O levels (review comment)	Lines removed; axis labelled "not a redox potential"; $-\varepsilon - 4.44$ columns renamed `*_qualitative`
Eosin Y optimisation would not converge (displacement test on a flat torsion)	`--criteria flat` (same gradient thresholds, no displacement test) and `--start` to resume
Rhodamine B optimisation recorded about 19 h of wall time	Diagnosed (Section 7.1): the laptop slept for about 15.5 h of it. One step costs 6 min, two thirds of it the solvent model. Long runs to be kept awake (`caffeinate`) or moved to a cluster
PubChem stores salts or neutral forms, not the active ions	Stored ions kept; `pubchem_parent_cid` + `pubchem_cid_relation` document the mapping
Fragment pattern did not match eosin Y, which silently gave empty localisation values	Pattern fixed; the script now stops with an error instead of writing empty values

7.1 Where the compute time goes

The rhodamine B optimisation recorded 68 073 s (≈19 h) of elapsed time for 20 steps. To find out why, one optimisation step (SCF from scratch plus nuclear gradient, 10 threads) was rerun on the converged geometry, with and without the solvent model, and the macOS power log was checked for the period of the original run.

Rhodamine B^+, 64 atoms, 617 basis functions	ddCOSMO water (s)	gas phase (s)
SCF (12 and 13 cycles)	156	71
Nuclear gradient	196	44
One optimisation step	352	115

In the solvated SCF, the density-fitted Coulomb/exchange build and the exchange–correlation integration each take about 2.4 s per cycle; most of the remaining time per cycle is the solvent.

Key finding: Most of the 19 hours was the laptop asleep. The power log shows the Mac asleep for about 15.5 h of the 18.9 h window, interrupted only by minute-long maintenance wakes. At the measured cost of about 6 min per step, a 20-step optimisation needs about 2 h. Within a step, the ddCOSMO water model accounts for two thirds of the time (237 of 352 s), mostly in the gradient.

Concept: Wall time and CPU time. Wall time is elapsed clock time: it keeps running while a machine sleeps or waits for other jobs. CPU time adds up the time spent on every core. PySCF logs both for each component; for the exchange–correlation step here, 13.7 s of CPU time took 2.3 s of wall time, so about six cores were busy on average. A wall time far above the sum of the logged component times means the job was not running, not that the chemistry is expensive.

8 Status board and plan

Done	
Catalogue	16 organic photocatalysts, PubChem-checked, charge states documented
3D structures	Multi-conformer MMFF94s geometries for all 16; HTML viewer
DFT geometries	6 of 16 (3 rigid, 3 flexible/charged); D3(BJ) test re-optimisations for BP and TX
Orbitals	HOMO/LUMO for all 16 (step 3 rerun on the new geometries); localisation for EY^{2-} (DFT geometry) and Mes-Acr^{+}
Geometry check	Step 5 run: $\lvert\Delta\text{gap}\rvert \leq 0.03$ eV rigid, ≤ 0.15 eV flexible
Dispersion	D3/D4 working on the Mac; validated
Documentation	README and REPORT updated; code-architecture chart
Plan (proposed order)	
1. Geometries	DFT optimisation of the remaining 10 dyes, rose bengal and fluorescein first; then rerun steps 3 and 5. Run with sleep disabled or on a cluster; test whether gas-phase optimisation plus a solvated single point reproduces the gaps (about one third of the cost per step)
2. TDDFT	S_1/T_1 excitation energies, so computed transitions (not orbital gaps) are compared with absorption
3. Basis / dispersion	def2-TZVP single points; D3(BJ) re-optimisation of the flexible dyes
Scope	
Solid-state work	not covered by this summary

9 Questions worth raising

- The geometry error is up to 0.15 eV for flexible dyes. Should all remaining dyes be DFT-optimised, or only the flexible/charged ones?
- For TDDFT, which functional does the group prefer?
- Is there cluster time for the larger molecules? One rhodamine B optimisation step takes 6 min on the laptop (about 2 h per optimisation), and TDDFT and the larger dyes will cost considerably more.